Tsz Fei Wang

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Education

University of Waterloo | Honours Computer Science (Co-op) — Sept 2026 – May 2031 (expected)

Skills & Technologies

Languages & Frameworks: TypeScript, JavaScript, HTML/CSS, Python, Java, SQL, GLSL, React, Next.js, Three.js, Qt/QML
Tools: Git, Vercel, Firebase, Canvas 2D, WebGL2, Node.js, Docker, GitHub Actions, Playwright, Postman

Experience

Krembil Research Institute, World Community Grid — Mar 2026 – Jun 2026
Software Engineer Intern — *Toronto, ON*

- Built the frontend of Catrin's research data portal, giving scientists one place to search, filter and browse **~4 million** gene regulatory associations across **33,000 genes**.
- Unified results from **9 independent databases** into a single searchable interface, so researchers can query them as one dataset.
- Credited as a contributor on the Catrin manuscript (in preparation).

Code 4 Tomorrow — Apr 2021 – Present
Vice President, Web Development Project Lead — *Remote*

- Launched a **JavaScript** frameworks course and a new workshop series for a nonprofit coding school, growing enrollment by **50+ students**.
- Led web development projects and course operations across **20+ courses** serving **400+ students**.

William Lyon Mackenzie C.I. — Nov 2024 – Present
Web Developer — *Toronto, ON*

- Maintain the official website of a **1,400+ student** high school, publishing announcements and keeping every page current for students and staff.
- Migrated the school's content off a deactivated information portal onto the new site, working with the administration.

Projects

Still Waters | *JavaScript, HTML/CSS, WebGL2* — *Jul 2026 – Present*

- Designed and shipped an interactive koi-pond portfolio in **JavaScript**, with **WebGL2** water shaders and layered **Canvas 2D** animation of koi, foliage and ripples.
- Made the scene react to visitors: clicks and taps send ripples that scatter the koi, with matching sound effects and ambient audio.
- Kept animation smooth on HiDPI and touch screens by scaling to the device pixel ratio, pre-rendering to offscreen canvases and pausing when the tab is hidden; deployed on **Vercel**.

Physical Studio | *React, Three.js, Python* | ***Winner**, Hack the Valley 11* — *Oct 2026*

- Built the **React Three Fiber** viewer that turns generated hardware designs into animated, step-by-step 3D assembly guides.
- Engineered the **Python** MCP server behind it, letting an LLM design projects from a library of measured parts and check its own work for collisions.

Submerge | *QML, GLSL, Python* — *Oct 2026 – Present*

- Shipped a full KDE Plasma theme (wallpaper, panel style, login screen) in **QML** and **GLSL**, downloaded **~1,290 times** on the KDE Store.
- Built a live system HUD (CPU, GPU, memory, network, battery) and resolved a user-reported NVIDIA/Wayland rendering failure by porting the shaders to GLSL ES.

Awards & Honors

Gold & Silver Medalist, IT Systems Networking, Skills Ontario (Regional & Provincial) — Mar 2026 – May 2026
National Champion x2, 1st in Canada, Cybersecurity, $1,000 scholarship — May 2025, Mar 2026
National Finalist, 2nd in Canada, Cybersecurity, $1,600 scholarship — Mar 2025